

EXPERIMENT NO 1

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INPUT CODE

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#include <stdio.h>

void displayMenu();

int length(char[]);

void compare(char[], char[]);

void palindrom(char[]);

void substring(char[], char[]);

void copy(char[], char[]);

void reverse(char[]);

int i, j;

int main() {

    int choice;

    char str[50];

    char str2[50];

    int exit = 0;

    displayMenu();

    do {

        printf("ENTER CHOICE (7 for Display Menu) :\n");

        scanf("%d", &choice);

        if (!(choice == 7 || choice == 8)) {

            printf("Enter String: ");
```

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    scanf("%s", str);
}
switch (choice) {
    case 1:
        printf("Length of the String is %d\n\n", length(str));
        break;
    case 2:
        printf("Enter second String: ");
        scanf("%s", str2);
        copy(str, str2);
        break;
    case 3:
        reverse(str);
        break;
    case 4:
        palindrom(str);
        break;
    case 5:
        printf("Enter substring String: ");
        scanf("%s", str2);
        substring(str, str2);
        break;
    case 6:
        printf("Enter second String: ");
        scanf("%s", str2);
        compare(str, str2);
```

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        break;

    case 7:

        displayMenu();

        break;

    case 8:

        exit = 1;

        break;

    default:

        printf("Invalid choice! Try again.\n\n");

        break;

}

} while (exit == 0);

return 0;

}

void displayMenu() {

    printf("<-----Menu----->\n");

    printf("1. Length\n");

    printf("2. Copy String in another String\n");

    printf("3. Reverse String\n");

    printf("4. Check for Palindrome\n");

    printf("5. Check for Substring\n");

    printf("6. Compare Any String with another\n");

    printf("7. Display menu\n");

    printf("8. Exit\n");

}

int length(char a[]) {

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int len = 0;

while (a[len] != '\0') {

    len++;

}

return len;

}

void compare(char str[50], char str2[50]) {

    i = 0;

    while (str[i] != '\0' && str2[i] != '\0' && str[i] == str2[i]) {

        i++;

    }

    if (str[i] == str2[i]) {

        printf("Both strings are EQUAL.\n\n");

    } else if (str[i] > str2[i]) {

        printf("First string is GREATER than second string.\n\n");

    } else {

        printf("First string is SMALLER than second string.\n\n");

    }

}

void palindrom(char a[]) {

    int l = length(a);

    int flag = 0;

    for (i = 0; i < l / 2; i++) {

        if (a[i] != a[l - 1 - i]) {

            flag = 1;

            break;

        }

    }

}

```

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    }
}
if (flag == 0) {
    printf("The String is a Palindrome\n\n");
} else {
    printf("The String is not a Palindrome\n\n");
}
}

void substring(char a[50], char b[50]) {
    int al = length(a);
    int bl = length(b);
    int flag = 0;
    if (bl > al) {
        printf("The String B is NOT a substring of A\n\n");
        return;
    }
    for (i = 0; i <= al - bl; i++) {
        int match = 1;
        for (j = 0; j < bl; j++) {
            if (a[i + j] != b[j]) {
                match = 0;
                break;
            }
        }
        if (match == 1) {
            flag++;

```

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    }
}
if (flag == 0) {
    printf("The String B is not substring of A\n\n");
} else {
    printf("The String B is substring of A and it occurs %d times\n\n", flag);
}
}

void copy(char a[], char b[]) {
    int bl = length(b);
    for (i = 0; i < bl; i++) {
        a[i] = b[i];
    }
    a[bl] = '\0';
    printf("The given string is Copied. The main String is: %s\n\n", a);
}

void reverse(char a[]) {
    int al = length(a);
    char temp;
    for (i = 0; i < al / 2; i++) {
        temp = a[i];
        a[i] = a[al - 1 - i];
        a[al - 1 - i] = temp;
    }
    printf("The reversed string is: %s\n\n", a);
}

```

OUTPUT

```
<<-----Menu----->>
1. Length
2. Copy String in another String
3. Reverse String
4. Check for Palindrome
5. Check for Substring
6. Compare Any String with another
7. Display menu
8. Exit
ENTER CHOICE (7 for Display Menu) :
1
Enter String: AMOGH
Length of the String is 5

ENTER CHOICE (7 for Display Menu) :
2
Enter String: AMOGH
Enter second String: NAMAN
The given string is Copied. The main String is: NAMAN

ENTER CHOICE (7 for Display Menu) :
3
Enter String: RAJU
The reversed string is: UJAR

ENTER CHOICE (7 for Display Menu) :
4
Enter String: NAMAN
The String is a Palindrome

ENTER CHOICE (7 for Display Menu) :
5
Enter String: KARNA
Enter substring String: KAR
The String B is substring of A and it occurs 1 times
```

```
<<-----Menu----->>
1. Length
2. Copy String in another String
3. Reverse String
4. Check for Palindrome
5. Check for Substring
6. Compare Any String with another
7. Display menu
8. Exit
ENTER CHOICE (7 for Display Menu) :
6
Enter String: DEHU
Enter second String: DAMAN
First string is GREATER than second string

ENTER CHOICE (7 for Display Menu) :
7
<<-----Menu----->>
1. Length
2. Copy String in another String
3. Reverse String
4. Check for Palindrome
5. Check for Substring
6. Compare Any String with another
7. Display menu
8. Exit
ENTER CHOICE (7 for Display Menu) :
8

...Program finished with exit code 0
Press ENTER to exit console. 
```